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DETAIL SPECIFICATION

DETECTOR, COMBINED CONTAMINATED FUEL

This specification is approved for use by the Naval Air Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for the combined contaminated fuel detector (CCFD). This specification is not applicable to any +100 additized petroleum products. This specification primarily covers combined contaminated fuel detectors with motorized pump and digital direct-reading free water detection capability. This specification does not cover units that utilize particle counting technology.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to Commanding Officer, Naval Air Warfare Center Aircraft Division Lakehurst, System Standardization and PHS&T Branch, Code BL32600, Mail Stop 120-3, Route 547, Joint Base MDL, NJ 08733-5100 or emailed to NAVAIR-Standardization@us.navy.mil.Since contact information can change, you may want to verify the currency of this address information using the ASSIST online database at https://assist.dla.mil.

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2.2 Government documents.

INTERNATIONAL STANDARDIZATION AGREEMENTS

DEF STAN 91-91	Defense Standard 91-91: Turbine Fuel, Kerosene Type, Jet A-1 (NATO Code F-35, Joint Service Designation: AVTUR)
NATO AFLP-3747	Guide Specifications (Minimum Quality Standards) for Aviation Turbine Fuels (F-24, F-27, F-34, F-35, F-37, F-40, and F-44)
NATO STANAG 1385	Guide Specification (Minimum Quality Standards) for Naval Distillate Fuels (F-75 and F-76)

(Copies of these documents are available from at https://www.nato.int.)

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-5624	Turbine Fuel, Aviation, Grade JP-5 (NATO F-44)
MIL-DTL-16884	Fuel, Naval Distillate
MIL-DTL-83133	Turbine Fuel, Aviation, Kerosene Type, JP-8 (NATO F-34) and NATO F-35

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-1399-300-1	Low Voltage Electric Power, Alternating Current
MS35058	Switch, Toggle One Pole, Sealed Toggle
MS35059	Switch, Toggle, Two Pole, Sealed Toggle

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

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ASTM INTERNATIONAL

ASTM D1655	Standard Specification for Aviation Turbine Fuels
ASTM D2276	Standard Test Method for Particulate Contaminant in Aviation Fuel by Line Sampling
ASTM D3240	Standard Test Method for Undissolved Water in Aviation Turbine Fuels

(Copies of these documents are available from https://www.astm.org.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 8217	Products from petroleum, synthetic and renewable sources - Fuels (class F) - Specifications of marine fuels
ISO 12103-1	Road Vehicles - Test Contaminants for Filter Evaluation - Part 1: Arizona Test Dust

(Copies of these documents are available from the at https://www.iso.org.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA Publication 70	National Electrical Code

(Copies of this document are available from at https://www.nfpa.org.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

ANSI/NEMA HP 3	Electrical and Electronic Polytetrafluoroethylene Insulated High-Temperature Hook-Up Wire, Types ET (250 V), E (600 V), and EE (1,000 V)

(Copies of this document are available from at https://www.nema.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.(See 6.3)

3.2 Materials. Detectors shall be made from such materials and by such processes as to ensure compliance with the requirements of this specification and the lightest practical weight when examined in accordance with 4.4.2.

3.2.1 Metals. Metals used in the manufacture of the detectors covered by this specification shall be resistant to corrosion due to fuels, salt spray or atmospheric conditions when in storage or during normal service life.

3.2.2 Fungus proof materials. Materials that are nutrients for fungi shall not be used in the detectors.

3.3 Construction. The construction of the CCFD shall meet the requirements of 3.3.1 through 3.3.13.11 when tested and examined in accordance with 4.4.2.

3.3.1 (CCFD) general design. The CCFD shall be a portable unit capable of determining the amount of particulate contamination in a fuel sample using light transmission between membrane filters, and free (undissolved) water by digital direct-reading free water detection technology. The vacuum pump shall be repairable at the first or second echelon repair levels. The detector shall be capable of translating particulate contamination results using 47 millimeter (mm) filters and free water contamination results using 25 mm test pads. The detector shall be capable of being hand-carried by personnel to a fueling site for the purpose of analyzing fuel for particulate contamination or water content. The detectors shall withstand the normal strains, jars, and any other conditions incident to service use. The detector shall be compatible and safe for use with middle distillate fuels (JP-5, JP-8, Jet A, Jet A-1, F-24, F-76, F-75, ISO 8217 DMA grade Marine Gas Oil).

3.3.2 Standalone digital free water detector (DFWD). The standalone DFWD unit shall be in accordance with ASTM D3240.

3.3.3 Weight. The weight of the completely assembled CCFD with attachments, components and case, shall be not greater than 27.0 kilograms.

3.3.4 Dimensions.

3.3.4.1 External dimensions. The maximum external dimensions of the detector shall be as follows:

Height	0.5 meter (19.7 inches)
Width	0.5 meter (19.7 inches)
Depth	0.3 meter (11.8 inches)

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3.3.4.2 Space or enclosure maximum dimensions of detector operation. The detector shall be capable of being operated in a space or enclosure of the following maximum dimensions:

Height	0.8 meter (31.5 inches)
Width	0.7 meter (29 inches)
Depth	0.6 meter (23.6 inches)

3.3.4.3 External dimensions of the standalone DFWD. The maximum external dimensions of the free water detector shall be as follows:

Height	0.09 meter (3.5 inches)
Width	0.28 meter (11 inches)
Depth	0.11 meter (4.25 inches)

3.3.5 Storage. The detector shall have a drawer, compartment or other means for storage of accessory items including a sampling bottle, wash bottle, bottle receiver assemblies (47 mm and 25 mm), tygon plastic tubing, membrane filters, free water test pads, forceps, calibration (Wratten or equivalent) filters, filter shuttles, verification shuttles, spare fuses and lamps. All accessory items shall conform to the requirements in 3.3.13. The storage means shall be readily accessible to operating personnel and shall not interfere with operation of the detector.

3.3.6 Filter receptacle/photocell/light detection housing. The filter receptacle tray used for particulate detection shall be 48.0+0.15 millimeters to accommodate 47 mm diameter membrane filters. The filter receptacle tray used for free water detection shall be of sufficient size to accommodate a 25 mm free water test pad. The tray shall be of such design as to slide into a fixed photocell housing and shall not bind at any point between the open and closed positions in the photocell housing. A small cutout shall be provided on the top surface of the tray/shuttle to facilitate placement and removal of the membrane filters. The photocell shall be a Vactec VTS-7070A or equivalent as approved by the procuring activity and shall be fixed securely in the housing. A frosted glass, which equally diffuses light across its surface area, shall be mounted between the filter receptacle tray and incandescent light, and shall be sealed with an O-ring or other means to prevent fuel from leaking into the detector case. The light detection unit shall be hermetically sealed against moisture and contact with fuel.

3.3.7 Bottle receiver assembly. The bottle receiver assembly shall consist of a bottle receiver, membrane filter holder, one-hole rubber stopper (size 8), and a ground wire with plug. The bottle receiver shall be the upper portion of the assembly properly sized so the bottle receiver shall fit over the neck of the sample bottle prior to inversion for the vacuum filtration operation. The membrane filter holder is the lower half of the assembly upon which the membrane filters are placed for filtration, and which shall mate with the mouth of the vacuum flask using a one-hole rubber stopper through which the lower tube of the filter holder is inserted. The top and bottom halves of the bottle receiver assembly shall fit together snugly with a twisting motion to secure the membrane filters during the filtration operation. The membrane filter holder shall create a complete seal and uniformly distribute fuel across the membrane filter. The bottle receiver assembly shall be capable of holding both a 25 mm and 47 mm pad. This may require the use of multiple assemblies or adapters (see 3.4.4). The assembly for free water detection shall have an orifice no greater than

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1 mm, in accordance with ASTM D3240.

3.3.7.1 Approved free water test pad holder. The detector shall be compatible with the free water test pad holder (NSN: 6665-01-642-9295) that has been approved for use in the fleet. The approved free water test pad holder shall mate with the mouth of the vacuum flask and create a complete seal.

3.3.8 Switches. Toggle type switches in accordance with MS35058 and/or MS35059 shall be provided to control the particulate and free water detection light sources, and to turn the vacuum pump on and off.

3.3.9 Free water test pad. The free water test pad shall be in accordance with ASTM D3240.

3.3.9.1 ASTM D3240 compliance. The digital free water detector shall be constructed to comply with all provisions of ASTM D3240 and shall report free water values digitally. The detector shall be designed to accommodate the required 25 mm test pad. All accessory items required to accommodate the 25 mm test pad shall be provided with the detector.

3.3.10 Vacuum receiving vessel. The vacuum receiving vessel shall be large enough to contain at least 1200 milliliters (mL) of filtered fuel at one time. The drain line of the receiver shall permit complete drainage by gravity of the receiving vessel. The receiving vessel shall be fitted with a high-level detector which shall prevent the vessel from being overfilled and allowing fuel to reach the vacuum pump.

3.3.11 Electrical.

3.3.11.1 Power requirements. The detector shall be designed to operate on ungrounded 115 volts AC, 60 Hz and type I shipboard power as specified in MIL-STD-1399-300-1.

3.3.11.2 Wiring. The external and internal wiring shall be orderly and assembled in such a manner that individual wires may be easily traced. Connections between components shall be made using Molex type connectors to the maximum extent possible for ease of field replacement. Screw terminals and soldered joints are also acceptable. Crimp-on solderless connections shall not be used. The wire used shall be in accordance with ANSI/NEMA HP 3.

3.3.11.3 Electrical components safety. Voltage and/or amperage regulating devices, transformers and capacitors shall be sealed with a fuel resistant material to prevent contact of fuel or vapors with internal components of the devices. All other electrical components and connections shall be insulated to protect against shock to personnel.

3.3.11.4 Motor. The motor shall be an explosion proof type with a current limiting device to prevent overheating in the event of pump malfunction. The motor shall comply with the NFPA Publication 70, Article 500 for Group D, Class I, Division 1 locations.

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3.3.11.5 Display meter. The display meter for particulate readings shall be capable of being read, without interpolation, in increments of 0.001 milliamperes between 0.500 milliamperes and one milliampere.

3.3.11.6 Power cord. The external power supply cord shall be a minimum 18-3 grounded cable, at least 5 feet long, with a male "U" ground attachment. The cord shall be capable of being removed from the cabinet for internal storage.

3.3.11.7 Circuit protection. Circuit protection shall consist of panel mounted fuses, which can be replaced from the exterior of the cabinet or mechanical type circuit breakers, which can be reset from the exterior of the cabinet.

3.3.11.8 Vacuum pump safety interlock. An electrical circuit shall be incorporated into the unit which shall shut off the vacuum pump and illuminate an indicator lamp labeled "vacuum flask overfill" on the detector's top panel whenever the high-level detector in the vacuum receiving vessel is actuated by overfilling the vessel. This interlock circuit shall automatically deactivate whenever fuel is drained from the receiving vessel below the level of the overfill detector. An additional interlock shall ensure the drain valve is closed before restarting the vacuum pump.

3.3.12 Finish. Unless otherwise specified by the acquiring activity (see 6.2), painting and finishing of the detectors shall be in accordance with the supplier's standard practice, provided the following minimum criteria are met or exceeded: All surfaces to be painted shall have been cleaned to be free of all foreign matter detrimental to painting. At least one coat of primer and one topcoat or equivalent shall be applied.

3.3.12.1 Color. Unless otherwise specified by the acquiring activity (see 6.2), the detectors shall be furnished in the manufacturer's standard color.

3.3.13 Accessory items.

3.3.13.1 General. All components necessary for filtration and measuring of particulate and free water concentrations shall be provided with each detector. Compartments, holders, and/or a drawer shall be provided for storage of accessories as specified in 3.3.5.

3.3.13.2 Sample bottle. A sample bottle with a one-liter capacity shall be provided. The bottle shall be constructed of a shatterproof material which shall be resistant to middle distillate fuels. The mouth of the bottle shall be 1 inch in diameter capable of being inserted into the mouth of the vacuum receiver without leakage and shall also be provided with a screw type cap. Lines shall be permanently marked on the outside of the bottle at the 200 milliliter, 500 milliliter, and 800 milliliter fill levels. Two small notches shall be filed in the top rim of the bottle or similar means shall be provided to allow air to enter the bottle during vacuum filtration.

3.3.13.3 Wash bottle. A wash bottle with a 0.5-liter capacity shall be provided. The bottle shall be constructed of a flexible shatterproof material which shall be resistant to middle distillate fuels. A fixed dispensing tube type cap shall be included with the wash bottle.

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3.3.13.4 Drain tubing. The drain tubing shall be middle distillate fuel compatible tygon type tubing or equivalent as approved by the procuring activity. The drain tubing shall be 3/8 inch in diameter, sized to fit securely on the barbed drain fitting, and shall be at least 4 feet in length. The drain tubing shall allow the vacuum receiving vessel to empty completely within 1 minute.

3.3.13.5 Membrane filters. Each detector shall be equipped with 100 membrane filters with a 47-millimeter diameter and 0.65-micron pore size (NSN 6635-00-967-0500 or Millipore Corporation catalog number DAWP 04700 or equivalent as approved by the procuring activity).

3.3.13.6 Free water test pads. Each detector shall be equipped with 100 free water detector pads (25 mm) in accordance with ASTM D3240 (NSN 6665-00-235-3820).

3.3.13.7 Particulate verification materials. Wratten filters or equivalent as approved by the procuring activity shall be provided in an individual covering to prevent prolonged, direct exposure to sunlight.

3.3.13.7.1 Calibration curve. Each detector shall have its own calibration curve. This curve shall show the relationship between the difference in the detector milliammeter reading and the actual amount of particulate contaminant in the fuel over the entire scale, expressed in milligrams/liter (mg/L). The calibration curve shall be checked in accordance with 4.4.8.

3.3.13.7.2 Wratten filters. A set of calibrated Wratten or Wratten 2 filters shall be provided with each detector. Wratten filters shall be 47 mm diameter, neutral density (ND) filters, one each with ratings at ND-0.10 and ND-0.20. The filters shall be individually wrapped and labeled as a set with the corresponding equivalent contaminant concentration.

3.3.13.8 Free water verification. Free water calibration shall be verified by one of the following methods.

3.3.13.8.1 Verification pads. Two (2) free water verification pads shall be provided with each detector. One verification pad shall provide an output value less than 5 ppm and one verification pad shall provide an output value greater than 15 ppm. The pads shall be clearly labeled with the respective free water concentration and markings to indicate correct orientation if needed. The pads shall be provided in individual coverings to prevent prolonged, direct exposure to sunlight.

3.3.13.8.2 Free water verification shuttle. A standard shuttle shall be provided with each detector. The standard shuttle contains a single fluorescent standard held within a shuttle holder. The shuttle holder has two apertures to the standard on opposite sides, the smaller aperture shall represent a value less than 5 PPM and the larger aperture a PPM value near the maximum detection limit value of the instrument. The shuttle shall be clearly labeled with the respective free water concentration represented by the standard and markings to indicate correct orientation. The shuttle shall be provided in an individual covering to prevent prolonged, direct exposure to sunlight.

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3.3.13.8.3 Internal calibration. A checking (or "fluorescing") standard shall be provided if the detector is self-calibrating in addition to a built-in calibrating standard. The checking standard shall be made to indicate a mid-rage reading, 4-6 ppm. A self-calibrating detector shall calibrate the unit with every single test and internally record time, date, ppm reading and temperature.

3.3.13.9 Miscellaneous. Each detector shall also be equipped with one pair of flat tipped forceps, 1 number 8 rubber stopper, 1 spare incandescent (or LED) lamp, 1 spare frosted glass and 2 spare fuses (except when mechanical circuit breakers are employed).

3.3.13.10 Identification. The switches, viewers, accessory slots, and other functional external parts of the detectors shall be legibly and durably marked in accordance with MIL-STD-130.

3.3.13.11 Workmanship. The detectors, including all parts and accessories, shall be fabricated and finished in a workmanlike manner and shall be free from blemishes, defects, burrs, sharp edges and incorrect dimensions and markings of parts and assemblies.

3.4 Performance.

3.4.1 Sample type. Devices shall meet all the requirements of this specification when any middle distillate fuel is chosen as the sample. For testing, the referee fuel to be used shall be JP-5 fuel conforming to MIL-DTL-5624.

3.4.2 Light source. The CCFD light source shall meet the requirements specified in 3.4.2.1 and 3.4.2.2 when tested in accordance with 4.4.3.

3.4.2.1 Light bulbs. The light bulbs for particulate detection shall be easily accessible for field replacement without the use of tools.

3.4.2.2 Light intensity. The intensity of the incandescent or LED light for particulate testing shall be mechanically or electrically adjustable from the exterior of the cabinet top to provide a meter reading over the range of 0.500 to 1.000 milliampere.

3.4.3 Voltage/frequency response.

3.4.3.1 Sediment detection. After a warm-up period of three minutes, when set at a meter reading of 0.600, the meter reading shall not drift by more than 0.005 milliampere in a period of three minutes and shall not fluctuate by more than 0.002 milliampere within a period of three minutes when tested in accordance with 4.4.4.1.

3.4.3.2 Free water detection. After a warm-up period of three minutes, with the lower value free water verification shuttle inserted, the meter reading shall not drift by more than 0.3 ppm in a period of three minutes and shall not fluctuate by more than 0.1 ppm within a period of three minutes when tested in accordance with 4.4.4.2 The lower value free water verification shuttle shall be reinserted before each of the nine trials.

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3.4.4 Bottle receiver assembly. The bottle receiver assembly shall be capable of preparing membrane filters and free water test pads that are representative of the fuel sample.

3.4.4.1 Membrane filter holder. The membrane filter holder shall evenly distribute fuel across the exposed area of the membrane filter. The radial distance of the 47 mm pad not exposed to fuel (i.e. surface area covered by top and bottom halves of the filter assembly) shall not exceed 5.0 mm from the circumference of the pad. The membrane filter holder shall be checked in accordance with 4.4.5.1.

3.4.4.2 Free water test pad holder. The free water test pad holder shall evenly distribute fuel across the exposed area of the free water test pad. The radial distance of the 25 mm pad not exposed to fuel (for example, surface area covered by top and bottom halves of the filter assembly) shall not exceed 2.5 mm from the circumference of the pad. The free water test pad holder shall be checked in accordance with 4.4.5.2.

3.4.5 Filtration time. The filtration system of the detector shall be capable of filtering a volume of the referee fuel through the membrane filter holder and free water test pad holder in accordance with the following requirements.

3.4.5.1 Membrane filter holder. The filtration system of the detector shall be capable of filtering an 800-milliliter sample of the referee fuel through two stacked 0.65-micron membrane filters in a maximum of two and one-half minutes using a minimum vacuum of 25 in/Hg (635 mm/Hg) minutes when tested in accordance with 4.4.6.

3.4.5.2 Free water test pad holder. The filtration system of the detector shall meet the filtration requirements in accordance with ASTM D3240.

3.4.6 Vacuum pump. The vacuum pump shall be capable of producing a minimum vacuum of 23 in/Hg (584.2 mm/Hg) and the pump discharge shall be to the outside of the detector case when tested in accordance with 4.4.7.

3.4.6.1 Vacuum pump safety interlock. An electrical circuit shall be incorporated into the unit which shall shut off the vacuum pump and indicate vacuum flask overfill on the detector's top panel whenever the high-level detector in the vacuum receiving vessel is actuated by overfilling the vessel. This interlock circuit shall automatically deactivate whenever fuel is drained from the receiving vessel below the level of the overfill detector. An additional interlock shall ensure the drain valve is closed before restarting the vacuum pump.

3.4.7 Particulate verification materials.

3.4.7.1 Calibration curve. Each detector shall have its own calibration curve. The calibration curve shall be checked in accordance with 4.4.8.

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3.4.7.2 Wratten filters. All Wratten filters shall be calibrated by the detector manufacturer in accordance with 4.4.7 and shall be assigned a number equivalent to a given contamination level. The Wratten filters shall be checked in accordance with 4.4.8.

3.4.8 Free water detection.

3.4.8.1 Free water detection accuracy, repeatability, and reproducibility. The manufacturer shall produce a round robin study in accordance with the Naval Fuels and Lubricants Standard Work Package SWP 44FL-027 to demonstrate the unit's repeatability, reproducibility, and accuracy of the digital free water detector. The round robin study may be waived if the free water detector is listed in ASTM D3240 with an ASTM defined repeatability and reproducibility. The detector shall be designed to accommodate the required 25 mm test pad. All accessory items (see 3.3.13) required to accommodate the 25 mm test pad shall be provided with the detector.

3.4.9 Free water verification materials. Detectors shall comply with all provisions of ASTM D3240, Calibrating Standard Section when tested as specified in 4.4.9.

3.4.10 Shock test. The unit shall undergo shock testing in accordance with 4.4.10. At the completion of testing, the unit shall still perform its intended functions.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection of the detectors shall consist of all the required examinations, inspections and tests specified in table I. (see 6.2 and 6.3)

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TABLE I. Examinations and inspections.

Characteristic	Requirement Paragraph	Verification Paragraph	First Article
Materials	3.2	4.4.2	X
Construction	3.3	4.4.2	X
Light source	3.4.2	4.4.3	X
Voltage/frequency response: Sediment detection stability Free water detection stability	 3.4.3.1 3.4.3.2	 4.4.4.1 4.4.4.2	X
Bottle receiver assembly: Membrane filter holder Free water test pad holder	 3.4.4.1 3.4.4.2	 4.4.5.1 4.5.5.2	X
Filtration time: Membrane filter holder Free water test pad holder	 3.4.5.1 3.4.5.2	 4.4.6 ASTM D3240	X
Vacuum pump	3.4.6	4.4.7	X
Particulate verification materials	3.4.7	4.4.8	X
Free water detection	3.4.8	4.4.9.1	X
Free water verification materials	3.4.9	4.4.9.2	X
Shock test	3.4.10	4.4.10	X

4.2.1 First article sample. The first article sample shall be one (l) complete unit for test. The first article sample shall be a representative production item. The sample shall be plainly identified by a securely attached durably marked tag containing the following information:

FIRST ARTICLE SAMPLE

DETECTOR, Combined Contaminated Fuel Manufacturer's Designation or Number
Name of Manufacturer
Part Number
Submitted by (Name) (Date) for First Article Inspection in accordance with the requirements of MIL-DTL-22612E(AS) under authorization (reference letter authorizing test)

4.3 Conformance inspection.

4.3.2 End item. Each lot of detectors shall be subjected to the applicable inspections and tests as specified in table I. Any defect or test failure shall be cause to reject the individual detectors (see 6.5).

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4.3.3 Visual inspection. Each detector shall be examined internally and externally to determine conformance with table II.

4.4 Test methods.

4.4.1 Sample type. Devices shall meet all of the test methods specified in 4.4 when any middle distillate fuel is chosen as the sample. For testing, the referee fuel to be used shall be JP-5 fuel conforming to MIL-DTL-5624. Prior to testing, the referee fuel shall contain less than 5 ppm free water and no more than 0.26 mg/L of solids when measured in accordance with ASTM D3240 and ASTM D2276, respectively.

4.4.2 Visual inspection. Each detector shall be examined internally and externally to determine conformance with table II and the requirements of this specification.

TABLE II. Visual inspections.

Defect
Unit missing required component parts.
Unit not free of cracks, tears, dents, or distortion.
Unit not free of rocking when placed on a flat surface.
Unit not free of sharp edges or burrs.
Component parts not properly secured to housing.
Components not properly sized.
Line cord not adequately secured within housing.
Line cord not free of cuts, tears, or abrasions.
Blades of plug (Line cord) not properly aligned.
Identification marking not complete, legible, or finish not permanent.
Free of scratches, chips, or blisters.
Unit not free of dirt, grease, or foreign matter.

4.4.3 Light source.

4.4.3.1 Light intensity. When the light intensity for particulate testing is adjusted to its highest value, a reading of 0.800 or higher shall be obtained on the display meter. When the adjustment is set to its lowest value, a reading of 0.500 milliampere or lower shall be obtained.

4.4.3.2 Free water detector. The free water detector portion of the unit shall be tested by operating the on/off switch and observing illumination of digital screen.

4.4.4 Voltage /frequency response.

4.4.4.1 Sediment detection stability. Connect the detector power cord to a variable voltage/frequency power supply. Turn the detector on and allow to warm up for 3 minutes at 115 VAC and 60 Hz. Adjust the display meter to an initial reading of 0.600 milliampere. Vary the

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voltage from 92 to 138 volts AC and the frequency from 56.7 to 63.4 Hz and record the display meter readings at each voltage/frequency setting shown in table III. The meter reading shall not drift by more than 0.005 milliampere or fluctuate by more than 0.002 milliampere from an initial setting of 0.600 milliampere.

4.4.4.2 Free water detection stability. Connect the detector power cord to a variable voltage/frequency power supply. Turn the detector on and allow to warm up for 3 minutes at 115 VAC and 60 Hz. Insert low reference free water standard shuttle. Vary voltage from 92 to 138 volts AC and the frequency from 56.7 to 63.4 Hz and record the display reading at each voltage/frequency setting shown in table III. The display reading shall not drift by more than 0.3 ppm from the initial display reading or fluctuate by more than 0.1 ppm.

TABLE III. Voltage/frequency test.

Voltage (volts AC)	Frequency (Hz)		
	56.7	60.0	63.4
92			
115			
138			

4.4.5 Bottle receiver assembly.

4.4.5.1 Membrane filter holder. The membrane filter holder shall evenly distribute fuel across the entire exposed area of the membrane filter. A fuel sample with a particulate concentration of 2.0 mg/L shall be prepared by adding 1.6 mg of ISO 12103-1 A1 ultrafine test dust to 800 mL of clean JP-5 fuel and mixing with an ultrasonic mixer for 30 seconds. The membrane filter preparation process shall consist of filtering the contaminated fuel through two 0.65-micron Millipore or equal filters. The Millipore filters shall be visually inspected for even distribution of dirt and shall be free of spotting. The radial distance of the 47 mm pad not exposed to fuel (i.e. surface is covered by top and bottom halves of the filter assembly) shall not exceed 5 mm from the circumference of the pad.

4.4.5.2 Free water test pad holder. The free water test pad holder shall evenly distribute fuel across the entire exposed area of the pad holder. A fuel sample with a free water concentration of 10 ppm shall be prepared by adding 5.0 µL of water to 500 mL of clean JP-5 fuel and mixing with an ultrasonic mixer for 30 seconds. The free water test pad preparation process shall consist of filtering the contaminated fuel through one free water test pad adhering to ASTM D3240. The free water test pad shall be visually inspected for even distribution of water and shall be free of spotting. The radial distance of the 25 mm pad not exposed to fuel (i.e. surface is covered by top and bottom halves of the filter assembly) shall not exceed 2.5 mm from the circumference of the pad.

4.4.6 Filtration time. The vacuum pump shall be sufficient to complete the filtration process in a maximum time of two- and one-half minutes. The filtration process shall consist of passing 800 mL of clean referee fuel through two 0.65-micron Millipore or equal filters. Clean referee fuel is fuel that has been filtered through 0.65-micron Millipore filters. The flow system shall be thoroughly examined to assure no leakage of fuel (see 3.4.6).

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4.4.7 Vacuum pump. Close the drain valve and insert a vacuum gage with rubber stopper into the vacuum flask mouth. Turn the pump switch on and record the maximum vacuum obtained. The acceptable vacuum is 25 in/Hg to 30 in/Hg (635 mm/Hg to 762 mm/Hg).

4.4.7.1 Vacuum flask overfill. With the 800 mL of fuel from 4.4.6 still in the vacuum flask, and the vacuum pump running, continue to add fuel to the flask until the overfill protection system is activated. Check that the pump has shut down, with no fuel carried over into the pump, and that the indicator lamp on top of the unit is illuminated. Open the drain valve and drain the fuel from the flask. The pump shall not restart until all the fuel is drained, and the drain valve returned to the closed position.

4.4.8 Calibration of particulate detector.

a. Turn the particulate light source switch to the "on" position and allow the light source to warm up for two to three minutes.

b. Thoroughly dry the filter receptacle before calibrating fuel, which may be on the receptacle surface, and could harm the Wratten filters.

c. Remove the silver foil wrapped Wratten calibration filters from their envelopes and unwrap them, leaving the filters exposed in the opened wrapper. The filter shall not be exposed any more than necessary to avoid coating the filter with dust. Forceps shall always be used to grasp the filters, and the area of contract shall be within ¼ inch of the outer edge of the filter to avoid damaging the effective surface.

d. Adjust the light intensity until a steady milliammeter reading of 0.600 is obtained.

e. Pick up one of the Wratten filters using the forceps (hold filters only within ¼ inch of outer edge), insert filter into the filter receptacle, slide the receptacle into the photocell housing, and record the milliammeter reading. A small chip or crack in the edge of the filter shall not affect the reading unless it extends into the illuminated part of the filter, which can be noted, when the filter is in the filter receptacle.

f. With forceps, remove the filter from the sliding receptacle and replace in opened wrappings. Replace the sliding receptacle and note milliammeter reading. If the milliammeter reading is not 0.600, readjust light intensity and repeat e.

g. Repeat e and f using the second Wratten filter.

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h. Unless duplicate readings are desired, the light switch shall be turned to the "off" position and the Wratten filters wrapped and returned to their envelopes.

i. A calibration plot shall be made in which the ordinate shall represent the difference in milliammeter readings from 0.0 to 0.13 milliamp with divisions at every 0.005 milliamp. The abscissa shall represent the amount of contaminant in a fuel from 0.0 to 10.0 milligrams per liter with divisions at every 0.2 milligram per liter and the equivalent milligrams per gallon with divisions at every 1.0 milligram per gallon.

j. To obtain the calibration point, subtract the lower of the two milliammeter readings of the Wratten filters from the higher reading. This difference (change in reading) is plotted on the calibration chart versus the weight of contaminant per liter given for the set of filters supplied. A second point is plotted at 0 milligram per liter and 0.01 milliamp change in light reading. Draw a straight line between the two points and extend the line to the edge of the chart. This line is the calibration line for the detector used.

4.4.9 Evaluation of free water detector performance and verification standards.

4.4.9.1 Free water detection accuracy, repeatability, and reproducibility. The manufacturer shall produce a round robin study in accordance with the Naval Fuels and Lubricants Standard Work Package SWP 44FL-027 or an ASTM inter-laboratory study (ILS) in accordance with ASTM D3240 to demonstrate the unit's repeatability, reproducibility, and accuracy of the digital free water detector.

4.4.9.2 Free water standards. The free water detector standard shall be tested in accordance with ASTM D3240.

4.4.10 Shock test. The unit shall be closed and lifted from the left side using the handle to a height of 6 inches and dropped onto a hard surface. Repeat for a total of 3 drops. Conduct the same procedure, lifting from the right side using the handle. After all six drops have been conducted, the unit shall still perform its intended functions.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The detectors covered by this specification are intended for use at Naval shore activities and aboard ships. The combined contaminated fuel detector is comprised of two portable instruments used to quickly determine the quantity of solid contamination and free water present in fuels. The detector processes a sample of a fuel on site in usually three to four minutes and establishes the level of contamination in the fuel. The level of contamination is measured by using the principle of differences in light transmission through membrane filters. For greater accuracy and to eliminate any fuel color effect, the two membrane filters are used in series. The first filter traps the solid contaminant, plus any fuel color effect. The second filter is subjected to only fuel color effects. The light transmission through the two filters is measured by a light sensitive cell system. Determining the difference between the amount of light transmitted through the contaminated and clean filters establishes the level of solid contamination in the fuel. The established level is evaluated with the calibration curve supplied with each detector to ascertain if the fuel is contaminated beyond required limits. All components necessary for filtration and measuring of transmitted light are incorporated into this one serviceable instrument. Also incorporated is a free water detector conforming to the requirements of the Naval Fuels and Lubricants Standard Work Package SWP 44FL-027 or ASTM D3240. A fuel sample is passed through a uranine-coated free water detector pad which is inserted in the pad holder; under the lamp illumination, free water absorbed by the pad fluoresces, and yields the fuel free water reading in parts per million. This product is known to produce invalid results with +100 additized fuels and therefore should not be used to determine particulate or free water content of +100 additized fuel. Due to the natural fluorescence of some diesel fuels, procedures for testing diesel fuels in these units may differ from aviation (kerosene) fuels.

6.1.1 MIL-DTL-22612D CCFDs. CCFDs approved to MIL-DTL-22612D are still valid for procurement under this version. This specification also provides details on the ASTM D3240 standalone digital free water detector.

6.1.2 ASTM D3240. This specification also provides details on the ASTM D3240 standalone digital free water detector.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Total quantity desired.
c. If first article is required (see 6.3).
d. Warning and special markings when required.
e. Finish and color other than manufacturer's standard (see 3.3.12)

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6.3 First article. When a first article is required for inspection and approval (see 3.1, 4.2, 6.2, 6.3.1and 6.3.2), the contract should specify the following provisions for first article inspection. When a contractor is in continuous production of these contaminated fuel detectors from contract to contract, consideration should be given to waive the first article inspections. If inspection is required, indicate:

a. That the first article inspection should be conducted at the Naval Air Warfare Center Aircraft Division (NAWCAD), with advance notification provided prior to shipping the first article test unit to NAWCAD.

b. That the approval of first article samples or the waiving of the first article inspection should not relieve the contractor of his obligation to fulfill all other requirements of the specification and contract.

6.3.1 First article test location. The first article test is to be conducted at the Naval Air Warfare Center, Aircraft Division in Patuxent River, MD. The center should be notified in writing of the contract award and estimated date of first article unit availability. The first article unit must be packaged securely and shipped to:

Naval Air Warfare Center
Naval Air Systems Command, AB44100
Building 2360, Propulsion Systems Evaluation Facility
22229 Elmer Road
Patuxent River, MD 20670-1534

6.3.2 Change approval. Any change in materials, construction, or performance of the detector must be approved by NAVAIR. Incorporation of any changes, which have not been so approved, must require retesting of the item in question.

6.4. Operational instructions. A set of instructions detailing the proper use, maintenance, and precautions of the unit must be provided to ensure safe operation.

6.5 Lot. A lot consists of all the detectors forming a part of one contract or order and submitted for acceptance at one time.

6.6 Subject term (key word) listing.

Free water
Instrument
Meter
Particle
Sediment

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

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Preparing activity:
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